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A Unified Framework to BRIDGE Complete and Incomplete Deep Multi-View Clustering under Non-IID Missing Patterns

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Abstract

Incomplete multi-view clustering (IMVC) has gained increasing attention due to its ability to analyze incomplete multi-view data. Despite deep IMVC methods achieved significant progress, they still face two challenges: (I) The method-specific inseparable designs limit their application. (II) Non-independent and identically distributed (Non-IID) missing patterns has not been considered and caused degeneration. To address these issues, we propose a novel unified framework that bridges from deep MVC to deep IMVC, while emphasizing the robustness against Non-IID missing patterns. Our framework has a two-stage process: (I) Multi-view learning on complete data, where our framework is modularly established to be compatible with different multi-view interaction objectives. (II) Transfer learning and clustering on incomplete data, where we propose a multi-view domain adversarial learning method to improve the model robustness to Non-IID missing patterns. Moreover, an intra-view and inter-view imputation strategy is introduced for more reliable clustering. Based on our unified framework, we easily construct multiple IMVC instances and extensive experiments verified their clustering effectiveness. The code is available at https://github.com/xiaorui-jiang/BRIDGE.

1. Introduction

Multi-view clustering (MVC) [8, 11, 26, 32, 44] has garnered increasing attention due to its ability to uncover the intrinsic cluster structures within multi-view data in an unsupervised manner. Traditional machine learning-based methods [16, 20, 25, 32, 35] and deep learning-based methods [7, 29, 40, 43, 47] constitute the two primary research paradigms in MVC. Behind them, a fundamental assumption is that each sample must be observed from multiple views and thus MVC can explore their common semantic information to improve clustering performance [20, 22, 27, 48]. For real-world application scenarios, however, some views of samples in multi-view datasets usually are missing or unusable, resulting in the development of incomplete multi-view clustering (IMVC) methods [18, 19, 28, 33, 45]. Owing to the superior representation capability of deep neural networks, deep-learning-based MVC methods (a.k.a., DMVC) and IMVC methods (a.k.a., DIMVC) have been receiving increasing attention. We summarize representative DMVC and DIMVC methods from recent literature as shown in Figure 1. Their primary techniques to achieve multi-view interaction could be categorized into mutual information-based, contrastive-based, and divergence-based approaches. Meanwhile, the key strategies to handle incomplete multi-view data include imputation-based and imputation-free methods.

Although extensive prior research has laid a solid foundation, several critical challenges remain to be addressed. ① **The method-specific nature of DIMVC limit its development and application.** Specifically, DIMVC methods often develop specialized network frameworks for training the data while incorporating the specialized modules to handle the incomplete multi-view data. However, the tight coupling between these components restricts the transferability and universality, preventing the seamless integration of effective strategies across different methods. For example, APADC [41] enhances complete multi-view data by maximizing multi-view mutual information while refining incomplete data representations through maximum mean discrepancy loss. Despite its effectiveness, this approach may face challenges in being directly adapted to alternative paradigms such as contrastive learning or deep divergence loss. This issue might limit the application potential of DIMVC, and thus we are encouraged to present a more flexible and method-agnostic solution, to enhance the adaptability of DMIVC methods and facilitate their effective comparison and integration. ② **Ignoring the potential risks of Non-IID missing patterns.** Existing IMVC methods typically follow the independent and identically distributed (IID) missing patterns assumption, which implies that complete and incomplete multi-view data share the same distribution.

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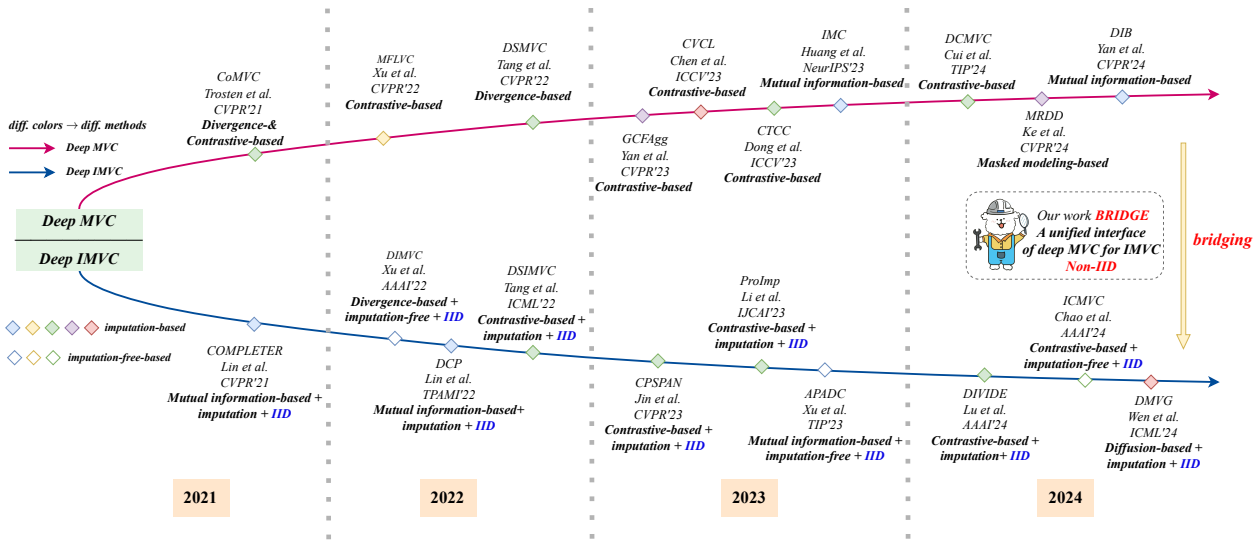


Figure 1. Our motivation is based on the following observations: (1) The training strategies for complete data in DIMVC closely resemble those in DMVC. However, the strong coupling of existing DIMVC methods has hindered the development of a modular and unified DIMVC framework; (2) Most DIMVC methods pay little attention to the challenges posed by the Non-IID missing patterns.

However, in real-world applications, unpredictably missing data is likely to cause distribution shifts between complete and incomplete multi-view data, and we call this case Non-IID missing patterns. For instance, sensor failures caused by light conditions might result in nonrandom noise which in turn perturbs the distribution of the collected data. Many IMVC methods (as illustrated in Figure 1) conduct data or representation recovery via the imputation of missing data, however, Non-IID missing patterns will introduce bias and greatly reduce the accuracy of imputation process. Some methods [37, 41] emphasize the importance of distributional or structural alignment in IMVC, yet they still operate under the IID missing patterns assumption. We observe that Non-IID missing patterns cause unignorable performance degeneration of many IMVC methods (please refer to the results in Figure 3). Therefore, addressing this issue is challenging and crucial as the missing data is usually structured and non-random for real-world complex scenarios.

To address the aforementioned issues, we propose *BRIDGE*, a novel modular DIMVC framework that decomposes IMVC into two relatively independent stages, i.e., I) multi-view learning on complete data, II) transfer learning and clustering on incomplete data. In the first stage, for achieving multi-view interaction, we observe that most DIMVC methods can adopt the model structure similar to DMVC, and they basically conduct multi-view representation learning on complete data. This inspires us to establish a framework that unifies the representation learning components of both DMVC and DIMVC, where the optimization objectives for multi-view interaction (e.g., mutual information-based [18, 41], contrastive-based [10, 43] and divergence-based [27, 29] losses) can be accommodated within our framework. This allows the basic models of DMVC to be reused by DIMVC, thereby enhancing their universality. In the second stage, we base on the data representations obtained from the first phase to conduct transfer learning and clustering on incomplete data. Specifically, we treat the complete and incomplete multi-view data as the source and target domains, respectively. During this stage, we employ self-training with clustering pseudo-labels and introduce domain adversarial constraints to enhance the model's robustness to Non-IID missing patterns. Subsequently, our method leverages aligned representations to recover the representations of missing data for final clustering. In summary, our main contributions include:

- We propose BRIDGE, a unified DIMVC framework which acts as an interface in methodology to enable DMVC methods to address IMVC problem. This new paradigm is hopeful to promote the development of both DMVC and IMVC, and also reduce their redundant work for the future.
- To the best of our knowledge, we are the first to consider Non-IID missing patterns in IMVC and incorporate domain adversarial training into the missing data imputation process, for alleviating the distribution discrepancy challenge. This simultaneously enhances the clustering performance in both IID and Non-IID missing patterns.
- We propose a modular solution for the DIMVC problem with an intra-view and inter-view imputation strategy, making BRIDGE have superior generality and compatibility. Based on it, we construct multiple DIMVC instances and extensive experiments validate their effectiveness.

Notations. Given a multi-view dataset with N samples, it has K classes and M views $\mathcal{X} = \{\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, ..., \mathbf{X}^{(M)}\}$, $\mathbf{X}^{(m)} \in \mathbb{R}^{N \times D_m}$, where D_m denotes the dimensionality of

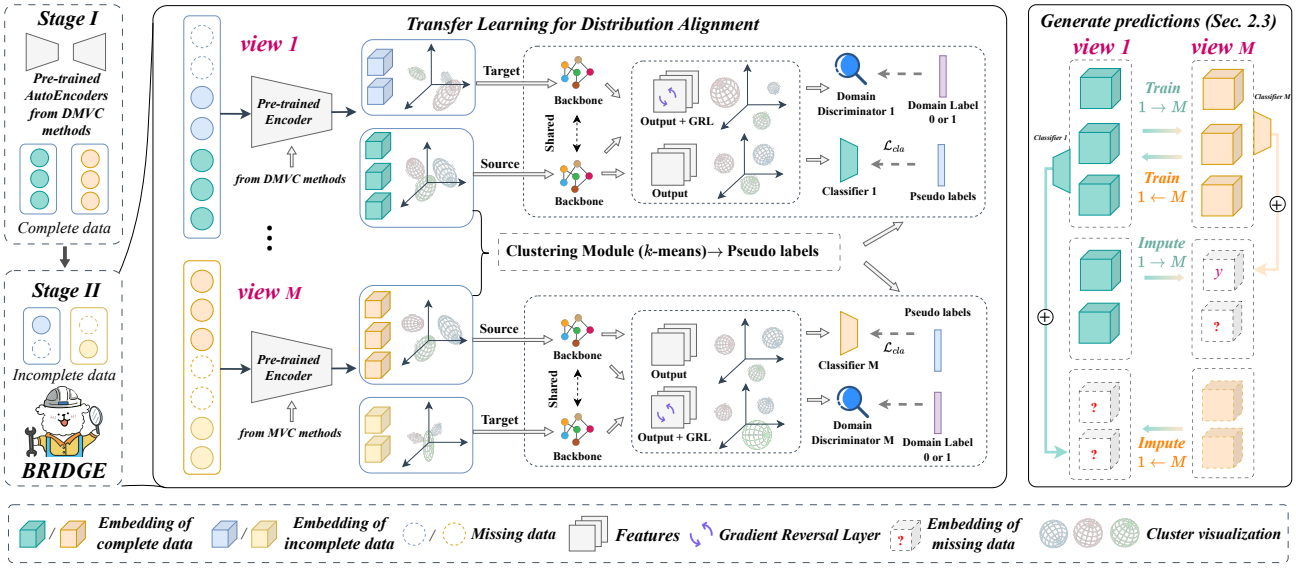


Figure 2. Our proposed BRIDGE, a unified framework designed to bridge DMVC and DIMVC. It consists of (1) multi-view learning on complete data (Sec. 2.1), (2) transfer learning for distribution alignment (Sec. 2.2), and (3) intra-view and inter-view imputation (Sec. 2.3).

samples in view m. At the same time, we consider the issue of missing instances: the data for view m is represented as $[\mathbf{X}_C^{(m)}; \mathbf{X}_I^{(m)}] \in \mathbb{R}^{N_m \times D_m}$, where $\mathbf{X}_C^{(m)}$ and $\mathbf{X}_I^{(m)}$ represent the complete data and incomplete data, while $N_m \leq N$ denotes the actual data volume, and the number of samples for complete data in each view is the same, denoted as n.

2. Method

2.1. Multi-View Learning on Complete Data

View-specific autoencoders are widely utilized in deep MVC and IMVC [12, 18, 30, 46] to extract view-specific information. For the purpose of establishing a unified framework, we also adopt autoencoders to extract information of each view. Specifically, an autoencoder pair is deployed for each view, denoted as $f_{\phi^m}^m(\cdot)$ and $g_{\psi^m}^m(\cdot)$ with learnable parameters ϕ^m and ψ^m, to perform feature representation and reconstruction. The encoder $f_{\phi^m}^m\left(\mathbf{X}^{(m)}; \phi^m\right) : \mathbf{X}^{(m)} \in \mathbb{R}^{N_m \times D_m} \longmapsto \mathbf{Z}^{(m)} \in \mathbb{R}^{N_m \times d_m}$ is responsible for nonlinearly mapping the raw data into a specific-dimensional space, while the decoder $g_{\psi^m}^m\left(\mathbf{Z}^{(m)}; \psi^m\right) : \mathbf{Z}^{(m)} \in \mathbb{R}^{N_m \times d_m} \longmapsto \hat{\mathbf{X}}^{(m)} \in \mathbb{R}^{N_m \times D_m}$ performs the reverse operation, where d_m is the dimensionality of embedded features. In IMVC, we have $\mathbf{X}^{(m)} = [\mathbf{X}_C^{(m)}; \mathbf{X}_I^{(m)}]$ and $\mathbf{Z}^{(m)} = [\mathbf{Z}_C^{(m)}; \mathbf{Z}_I^{(m)}]$. Generally, the reconstruction loss $\mathcal{L}_{recon}$ is achieved by:

$$\mathcal{L}_{recon} = \sum_{m=1}^{M} \left\| \mathbf{X}^{(m)} - \hat{\mathbf{X}}^{(m)} \right\|_F^2. \tag{1}$$

In the first stage, we aim to train on complete data using DMVC methods to equip the autoencoder with strong representation capabilities. Following a modular design principle, many representation learning-based DMVC methods [4, 13, 29, 40, 42–44] can be seamlessly integrated.

Specifically, the training methods for DMVC are highly diverse, with several classical approaches being widely recognized. These include mutual information-based [18, 24], contrastive-based [17, 43], and divergence-based [34, 39] methods. Thus, we integrate three training strategies for complete data into the fundamental BRIDGE framework. Specifically, we use I to denote the mutual information, $\mathbf{d}(\cdot, \cdot)$ as the similarity measurement function, τ_t as the temperature coefficient, D_{KL} as the Kullback-Leibler (KL) divergence, $\mathbf{P}$ as the target distribution (that is pseudo-labels for self-supervised learning obtained by integrating multiple views [42]), and $\mathbf{Q}_v$ as the cluster assignment distribution of view v. Taking a dual-view scenario (view 1 and view 2) as an example, the basic mutual information loss is:

$$\mathcal{L}_{mi} = -\sum_{i=1}^{n} I(\mathbf{z}_i^{(1)}, \mathbf{z}_i^{(2)}), \tag{2}$$

the basic contrastive loss is:

$$\mathcal{L}_{cl} = -\sum_{i=1}^{n} \log \frac{e^{\mathbf{d}(\mathbf{z}_i^{(1)}, \mathbf{z}_i^{(2)})/\tau_t}}{\sum_{j=1}^{n} \sum_{v=1,2} e^{\mathbf{d}(\mathbf{z}_i^{(1)}, \mathbf{z}_j^{(v)})/\tau_t}}, \tag{3}$$

and the basic KL divergence loss is:

$$\mathcal{L}_{dd} = D_{KL}(\mathbf{P} || \mathbf{Q}_v) = \sum_{i=1}^{n} \sum_{j=1}^{K} p_{ij} \log \frac{p_{ij}}{q_{ij}^v}. \tag{4}$$

In the literature, many DMVC and DIMVC methods leverage one kind of loss in $\{\mathcal{L}_{mi}, \mathcal{L}_{cl}, \mathcal{L}_{dd}\}$ to achieve the multi-view interaction while adopting the reconstruction loss $\mathcal{L}_{recon}$ in Eq. (1) to regularize the learning process. Detailed training strategies can be found in [18, 39, 43]. Although

previous methods typically share the common architecture to conduct multi-view learning, they usually have method-specific inseparable designs which heavily limit their mutual comparison and applications. In this paper, we integrate these approaches in our unified framework (Figure 2) and further build bridges from DMVC to DIMVC.

2.2. Transfer Learning for Distribution Alignment

After robust training on complete data, the model acquires strong representation capabilities. However, directly applying this model to incomplete data may lead to generalization errors, as it has not been trained on such data. In other words, using an encoder that has been carefully trained on complete data for incomplete data can result in a train-test discrepancy. Moreover, if the distribution of incomplete data differs from that of complete data (e.g., sensor failures introducing random and non-uniform noise into the incomplete data), this error may be further amplified.

To address this issue, a promising approach is to align the distributions between the representations of complete and incomplete data, thereby enabling the model trained on the former to generalize to the latter with minimal error. Therefore, we employ domain adversarial training [9, 31] to achieve this. Specificially, for view m, we treat the complete features $\mathbf{Z}_C^{(m)}$ (i.e., those derived from complete data $\mathbf{X}_C^{(m)}$) as the source domain and the incomplete features $\mathbf{Z}_I^{(m)}$ (i.e., those derived from incomplete data $\mathbf{X}_I^{(m)}$) as the target domain, employing a feature extractor $\mathcal{F}_{\eta^m}(\cdot)$ with parameters η^m to extract features. Next, we define a classifier $\mathcal{C}^{(m)}(\cdot)$ aimed at correctly distinguishing complete features. Remarkably, the training of the classifier is supervised (requiring labels for source domain data), which can be obtained by performing k-means clustering [23] on complete features.

$$\mathbf{Y}_C \leftarrow \mathbf{k\text{-}means}\left(\mathbf{fusion}\left(\{\mathbf{Z}_C^{(m)}\}_{m=1}^M\right), K\right). \tag{5}$$

Here, we use $\mathbf{Y}_C$ to represent the labels of complete features and $\mathbf{fusion}(\cdot)$ to denote the fusion function, which can have various implementations. For simplicity, we choose to concatenate multi-view data to obtain the fused result. Then we need to optimize the following classification loss:

$$\mathcal{L}_{\text{cls}}^{(m)} = \mathbb{E}_{\mathbf{Z}_C^{(m)}}\left[\ell\left(\mathcal{C}^{(m)}(\mathcal{F}_{\eta^m}(\mathbf{Z}_C^{(m)})), \mathbf{Y}_C\right)\right], \tag{6}$$

where $\ell(\cdot,\cdot)$ is the loss function for classification. Also, we define a domain discriminator $\mathcal{D}^{(m)}(\cdot)$, such that the output for $\mathcal{F}_{\eta^m}(\mathbf{Z}_C^{(m)})$ approaches 1 (source domain), while the output for $\mathcal{F}_{\eta^m}(\mathbf{Z}_I^{(m)})$ approaches 0 (target domain). We define $GRL(\cdot)$ as the Gradient Reversal Layer, which is used to invert the gradient signals in order to optimize the feature extractor to produce domain-invariant features [9]. Based on this, we formulate the following domain adversarial loss:

$$\begin{aligned}\mathcal{L}_{\text{adv}}^{(m)} = &-\mathbb{E}_{\mathbf{Z}_C^{(m)}}\left[\log \mathcal{D}^{(m)}(GRL(\mathcal{F}_{\eta^m}(\mathbf{Z}_C^{(m)})))\right] \\ &-\mathbb{E}_{\mathbf{Z}_I^{(m)}}\left[\log\left(1-\mathcal{D}^{(m)}(GRL(\mathcal{F}_{\eta^m}(\mathbf{Z}_I^{(m)})))\right)\right].\end{aligned} \tag{7}$$

The purpose of Eq. (7) is to confuse the representations of source domain and target domain within each view through a gradient reversal layer, thereby compelling the feature extractor to produce representations with as consistent a distribution as possible. On the other hand, the features of source domain are constrained by $\mathbf{Y}_C$ through the classifier, ensuring the accuracy of feature extraction. Jointly optimizing the classification loss and the adversarial loss facilitates domain adaptation, allowing the model trained on complete features to be effectively transferred to incomplete features. In other words, for view m, the following distribution alignment loss needs to be optimized (where γ is the weighting coefficient):

$$\mathcal{L}_{\text{DA}}^{(m)} = \mathcal{L}_{\text{cls}}^{(m)} + \gamma\mathcal{L}_{\text{adv}}^{(m)}. \tag{8}$$

2.3. Intra-&Inter-View Imputation for IMVC

After aligning the distributions of complete and incomplete features, the view-specific feature extractor reduces the bias caused by Non-IID missing patterns, laying the foundation for robustly predicting the features of missing data. In the most extreme case, where all other views in the multi-view data are missing and only a single view is available, BRIDGE can perform reliable missing value imputation through both intra-view and inter-view prediction.

As previously mentioned, we treat the predictions of missing data as a train-test task, training a well-performing model on complete data to transfer to incomplete data. Taking the imputation of view 1 and view m as an example, minimizing the conditional entropy $H(\mathcal{F}_{\eta^1}(\mathbf{Z}^{(1)})|\mathcal{F}_{\eta^m}(\mathbf{Z}^{(m)})$ can facilitate the recovery of data in view 1:

$$\min \mathbb{E}_{\mathcal{P}(\mathbf{H}^{(1)},\mathbf{H}^{(m)})}\|\Theta^{(m\to1)}(\mathbf{H}^{(m)}) - \mathbf{H}^{(1)}\|^2, \tag{9}$$

where $\mathbf{H}^{(m)} = \mathcal{F}_{\eta^m}(\mathbf{Z}^{(m)})$ and the model $\Theta^{(m\to1)}(\cdot)$ is used to learn the mapping from view m to view 1 [19]. In our framework, we first train model $\Theta^{(m\to1)}(\cdot)$ on features $\mathbf{H}_C^{(m)}$ (obtained by extracting complete features $\mathbf{Z}_C^{(m)}$ through the feature extractor $\mathcal{F}_{\eta^m}(\cdot)$) using the MSE loss:

$$\mathcal{L}_{pred}^{(m\to1)} = \|\Theta^{(m\to1)}(\mathbf{H}_C^{(m)}) - \mathbf{H}_C^{(1)}\|_F^2. \tag{10}$$

Then, for the j-th incomplete feature $\mathbf{z}_{I,j}^{(m)} \in \mathbf{Z}_I^{(m)}$ in view m, we first utilize the feature extractor within this view to obtain its distribution-aligned feature $\mathbf{h}_{I,j}^{(m)} = \mathcal{F}_{\eta^m}(\mathbf{z}_{I,j}^{(m)})$, so its predicted feature in view 1 can be represented as:

$$\hat{\mathbf{h}}_{P,j}^{(m\to1)} = \Theta^{(m\to1)}(\mathbf{h}_{I,j}^{(m)}). \tag{11}$$

However, we observe that relying solely on the inter-view predictions obtained from $\Theta^{(m\to1)}(\cdot)$ is insufficient. This

Table 1. Clustering performance gains (ACC%) compared with BRIDGE+MI (Part 1/2). $\mathcal{M}$ represents the missing rate, and the best results are highlighted in **bold**. Other metrics (NMI, ARI and PUR) show similar trends (the same applies to Table 2; please see the Appendix).

Method	BDGP			BBCSport			Hdigit			MNIST-USPS		
	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$
CDIMC-net	86.4-1.7	84.6+1.3	82.2+3.5	44.5+4.9	35.7-2.0	36.9+6.4	45.7-52.2	44.4-52.2	31.3-61.0	51.4-45.1	45.9-35.5	40.6-36.5
COMPLETER	41.0-47.1	42.8-40.5	42.2-36.5	33.3-6.3	36.2-1.5	35.9+5.4	86.4-11.5	97.6+1.0	95.7+3.4	96.9+0.4	93.8+12.4	80.8+3.7
DIMVC	80.3-7.8	81.7-1.6	65.3-13.4	31.6-8.0	35.5-2.2	30.0-0.5	97.2-0.7	96.3-0.3	95.4+3.1	81.1-15.4	80.0-1.4	74.9-2.2
DCP	48.0-40.1	50.3-33.0	31.2-47.5	37.7-1.9	37.3-0.4	27.8-2.7	86.7-11.2	98.1+1.5	96.5+4.2	65.0-31.5	76.4-5.0	55.8-21.3
DSIMVC	96.1+8.0	95.5+12.2	93.0+14.3	61.6+22.0	47.4+9.7	51.8+21.3	98.5+0.6	97.5+0.9	96.3+4.0	98.1+1.6	97.1+15.7	94.8+17.7
ProImp	94.6+6.5	84.2+0.9	84.8+6.1	50.4+10.8	49.3+11.6	42.3+11.8	98.1+0.2	97.8+1.2	81.3-11.0	92.9-3.6	89.2+7.8	88.7+11.6
APADC	79.7-8.4	75.3-8.0	65.4-13.3	38.4-1.2	37.3-0.4	36.8+6.3	43.2-54.7	40.3-56.3	63.9-28.4	97.2+0.7	95.3+13.9	92.1+15.0
CPSPAN	93.3+5.2	92.7+9.4	92.6+13.9	49.5+9.9	48.9+11.2	43.0+12.5	95.3-2.6	96.0-0.6	95.6+3.3	77.4-19.1	79.3-2.1	76.2-0.9
ICMVC	73.6-14.5	75.6-7.7	65.8-12.9	33.6-6.0	28.5-9.2	25.9-4.6	98.8+0.9	97.0+0.4	95.1+2.8	**98.6**+2.1	96.2+14.8	94.3+17.2
DIVIDE	92.6+4.5	89.2+5.9	74.6-4.1	25.4-14.2	26.3-11.4	26.5-4.0	97.7-0.2	96.3-0.3	93.6+1.3	95.2-1.3	88.1+6.7	89.6+12.5
BRIDGE+MI	88.1	83.3	78.7	39.6	37.7	30.5	97.9	96.6	92.3	96.5	81.4	77.1
BRIDGE+DD	95.7+7.6	93.6+10.3	90.2+11.5	**86.6**+47.0	**83.1**+45.4	**76.7**+46.2	93.9-4.0	93.1-3.5	91.1-1.2	86.5-10.0	78.4-3.0	73.5-3.6
BRIDGE+CL	**97.4**+9.3	**95.6**+12.3	**93.4**+14.7	83.5+43.9	72.6+34.9	76.0 +45.5	**99.0**+1.1	**98.0**+1.4	**97.0**+4.7	**98.6**+2.1	**97.4**+16.0	**95.3**+18.2

is because, when the missing rate is high, the reduction in training data limits the effectiveness of learning the mapping from view m to view 1. To address this issue, we propose fully leveraging intra-view information. Specifically, for $\mathbf{h}_{I,j}^{(m)}$, we obtain its predicted label using classifier $\mathcal{C}^{(m)}$:

$$y_{I,j}^{(m)} = \mathcal{C}^{(m)}(\mathbf{h}_{I,j}^{(m)}). \tag{12}$$

Since we have already determined the label of $\mathbf{h}_{I,j}^{(m)}$, we can then retrieve the distribution-aligned features in $\mathbf{H}_C^{(1)} = \mathcal{F}_{\eta^1}(\mathbf{Z}_C^{(1)})$ that share the same label and compute their average aggregation. As a reminder, the labels of $\mathbf{H}_C^{(1)}$ are obtained through Eq. (5). For the t-th feature $\mathbf{h}_{C,t}^{(1)} \in \mathbf{H}_C^{(1)}$ and its label $y_{C,t} \in \mathbf{Y}_C$, if $y_{C,t}$ matches $y_{I,j}^{(m)}$, it is included in the aggregation to obtain the predicted feature:

$$\bar{\mathbf{h}}_{P,j}^{(m\to 1)} = \frac{1}{\sum_t \mathbb{I}(y_{C,t} = y_{I,j}^{(m)})} \sum_t \mathbf{h}_{C,t}^{(1)} \, \mathbb{I}(y_{C,t} = y_{I,j}^{(m)}), \tag{13}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. Through this approach, we impute the missing value in view 1 using the information from view 1. This method yields a predicted value with strong fundamental reliability—owing to its averaging nature, it is less personalized but more robust—thereby enhancing the overall stability of the prediction results.

Then, by combining Eq. (11) and Eq. (13), we use λ as a weighting coefficient to derive the final imputation result:

$$\mathbf{h}_{P,j}^{(m\to 1)} = \lambda \hat{\mathbf{h}}_{P,j}^{(m\to 1)} + (1-\lambda)\bar{\mathbf{h}}_{P,j}^{(m\to 1)}. \tag{14}$$

For all features in $\mathbf{H}_I^{(m)}$, performing the above computation in parallel yields the entire imputation results.

Establish a unified clustering structure. After obtaining the predictions for all missing data, we construct the final feature matrix $\mathbf{H}$, which integrates complete, incomplete, and imputed features across all views. We use $\mathbf{H}_C^{(m)}$ and $\mathbf{H}_I^{(m)}$ to respectively represent the complete and incomplete features, and use $\mathbf{H}_P^{(o\to m)}$ to denote the imputed features for view m obtained from view o. Each row of $\mathbf{H}$ corresponds to a specific feature type (complete, incomplete, or imputed) and each column corresponds to a particular view. The detailed composition of $\mathbf{H}$ is:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_C^{(1)} & \mathbf{H}_C^{(2)} & \cdots & \mathbf{H}_C^{(M)} \\ \mathbf{H}_P^{(2\to 1)} & \mathbf{H}_I^{(2)} & \cdots & \mathbf{H}_P^{(2\to M)} \\ \mathbf{H}_I^{(1)} & \mathbf{H}_P^{(1\to 2)} & \cdots & \mathbf{H}_P^{(1\to M)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{H}_P^{(M\to 1)} & \mathbf{H}_P^{(M\to 2)} & \cdots & \mathbf{H}_I^{(M)} \end{bmatrix}_{N*M} \tag{15}$$

Finally, clustering on the final representation $\mathbf{H}$ yields the clustering predictions for all samples:

$$\mathbf{Y} \leftarrow \mathbf{k\text{-}means}(\mathbf{H}, K). \tag{16}$$

3. Experiments

3.1. Experimental Setup

Our methods. As a unified framework, our proposal is used to handle incomplete data. Therefore, we integrate BRIDGE into three common DMVC training methods, resulting in corresponding complete methods as presented below:

(1) BRIDGE + MI. Trains the complete data using the mutual information maximization approach in Eq. (2).

(2) BRIDGE + CL. Trains the complete data using the contrastive learning approach in Eq. (3).

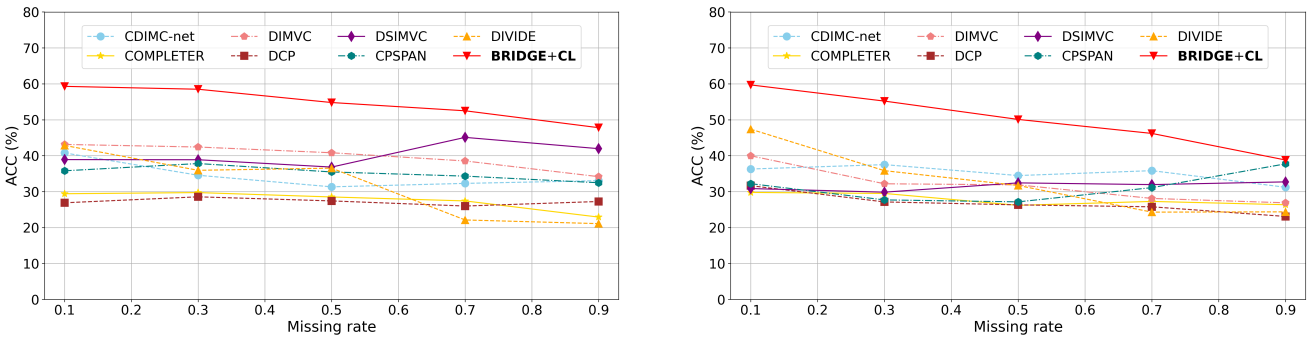
(3) BRIDGE + DD. Trains the complete data using the divergence-based approach in Eq. (4).

Comparison methods. We choose 10 classic and state-of-the-art deep IMVC methods for an in-depth comparison, including CDIMC-net [36], COMPLETER [18], DIMVC [39], DCP [19], DSIMVC [27], ProImp [15], APADC [41], CPSPAN [14], ICMVC [3], and DIVIDE [21].

Datasets. We conduct extensive comparisons and evaluations of the above methods on eight datasets, including BDGP [2], BBCSport [38], Hdigit [5], MNIST-USPS [25], Cifar10 [43], Cifar100 [43], Reuters [1] and NUSWIDE [6].

Table 2. Clustering performance gains (ACC%) compared with BRIDGE+MI (Part 2/2). $\mathcal{M}$ represents the missing rate, OOM denotes out of memory, and the best results are highlighted in **bold**. Only methods that support three or more views are compared.

Method	Cifar10			Cifar100			Reuters			NUSWIDE		
	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$	$\mathcal{M}=0.3$	$\mathcal{M}=0.5$	$\mathcal{M}=0.7$
CDIMC-net	OOM	OOM	OOM	OOM	OOM	OOM	17.3-16.9	17.1-15.7	16.8-14.5	32.6-7.0	30.1-5.7	26.3-5.3
COMPLETER	96.3-0.5	94.5+0.9	92.8+0.6	29.3+5.3	29.6-7.6	24.5+0.9	26.2-8.0	29.9-2.9	25.5-5.8	27.8-11.8	25.5-10.3	25.1-6.5
DIMVC	95.8-1.0	93.9+0.3	92.2+0.0	93.9+69.9	89.1+51.9	91.2+67.6	48.8+14.6	50.8+18.0	45.5+14.2	42.5+2.9	38.4+2.6	35.8+4.2
DCP	96.3-0.5	94.5+0.9	92.9+0.7	45.0+21.0	40.4+3.2	45.7+22.1	26.6-7.6	29.8-3.0	24.4-6.9	28.2-11.4	31.4-4.4	26.5-5.1
DSIMVC	95.4-1.4	92.4-1.2	87.3-4.9	21.4-2.6	17.1-20.1	23.1-0.5	48.6+14.4	44.2+11.4	44.8+13.5	37.9-1.7	47.9+12.1	42.5+10.9
CPSPAN	57.2-39.6	40.5-53.1	66.1-26.1	60.9+36.9	64.1+26.9	59.9+36.3	44.2+10.0	40.1+7.3	41.3+10.0	34.7-4.9	36.6+0.8	35.8+4.2
DIVIDE	81.8-15.0	77.6-16.0	73.4-18.8	93.3+69.3	89.6+52.4	84.8+61.2	27.6-6.6	29.5-3.3	27.4-3.9	36.5-3.1	30.8-5.0	26.1-5.5
BRIDGE+MI	**96.8**	93.6	92.2	24.0	37.2	23.6	34.2	32.8	31.3	39.6	35.8	31.6
BRIDGE+DD	95.8-1.0	94.1+0.5	90.9-1.3	91.9+67.9	92.7+55.5	92.8+69.2	**60.1**+25.9	**58.6**+25.8	**55.9**+24.6	49.8+10.2	49.9+14.1	48.3+16.7
BRIDGE+CL	96.6-0.2	**94.9**+1.3	**93.1**+0.9	**98.6**+74.6	**95.0**+57.8	**98.4**+74.8	50.7+16.5	52.2+19.4	46.2+14.9	**59.2**+19.6	**56.1**+20.3	**53.8**+22.2



(a) Class and quantity imbalance between complete and incomplete data.

(b) Scenario with random noise added to incomplete data.

Figure 3. Comparison results of different methods under Non-IID missing patterns on the NUSWIDE dataset.

Metrics. We use clustering accuracy (ACC), normalized mutual information (NMI), adjusted rand index (ARI), and purity (PUR) as evaluation metrics. Moreover, we also introduce a new metric specifically designed to evaluate the effectiveness of incomplete data processing modules in IMVC: the Final-to-Complete Accuracy Ratio (FCR), which is defined as the ratio of the final accuracy (i.e., ACC) to the accuracy on complete data (denoted as ACC_{cp}):

$$\text{FCR} = \frac{\text{ACC}}{\text{ACC}_{\text{cp}}}. \tag{17}$$

Compared to the traditional accuracy, FCR provides a more reliable evaluation of the effectiveness of incomplete data processing modules. This is because a model that performs well on complete data but is paired with a suboptimal incomplete data processing module may still achieve a high ACC, whereas its FCR would be relatively low, and vice versa.

3.2. Comparison Results

Comparison under IID missing patterns. Tables 1 and 2 present the clustering comparison results between our proposed methods and other methods under IID missing patterns. BRIDGE+MI is taken as the baseline method, and we report the increase or decrease in performance of other methods relative to it. An intuitive result is that our proposed three baseline algorithms achieve the best performance in almost all settings, whether for datasets with two views (Table 1) or more views (Table 2). But this lead is not dominated by a single method; instead, the three methods have varied successes. This fully shows the superior performance of our methods in conventional IID missing scenarios.

The significance of a flexible framework. Furthermore, when considering the performance of all methods across eight datasets, no single method consistently achieves the best results. For instance, DSIMVC performs well in most cases but experiences a notable decline on Cifar100; mutual information-based methods (e.g., COMPLETER, DCP, and BRIDGE+MI) perform poorly on BBCSport yet excel on Cifar10; BRIDGE+CL performs well in most scenarios but lacks a distinct advantage on Reuters. In other words, different methods inevitably exhibit variability across datasets, further emphasizing the importance of a flexible framework like BRIDGE, as real-world applications may require switching training strategies to adapt to different datasets.

Comparison under Non-IID missing patterns. We further compare the performance of different methods on NUSWIDE under two Non-IID missing patterns: class and quantity imbalance, as well as distribution shifts caused by random noise, as shown in Figure 3. It can be observed that our BRIDGE+CL consistently achieves the best performance across different Non-IID missing patterns, demonstrating the robustness of our method in handling potential distribution misalignment. More details can be found in the Appendix.

3.3. Ablation Study

In Table 3 and Table 4, we conduct ablation experiments on NUSWIDE and BDGP under both IID and Non-IID missing

Table 3. Ablation study under IID missing patterns (%).

Dataset	Components				BRIDGE + MI			BRIDGE + DD			BRIDGE + CL		
	(a)	(b)	(c)	(d)	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
NUSWIDE	✓		✓		35.0	9.6	7.5	41.3	12.3	10.0	41.7	14.2	12.7
	✓	✓			25.9	4.1	0.9	25.4	3.8	0.5	26.5	5.6	0.9
		✓	✓	✓	29.1	8.0	4.8	36.4	11.0	8.5	37.7	12.9	9.3
	✓	✓	✓		34.8	9.7	7.4	41.8	12.7	9.9	42.8	15.0	13.5
	✓	✓	✓	✓	36.3	10.4	8.3	42.5	12.9	10.3	44.2	15.7	14.3
BDGP	✓		✓		51.5	31.6	23.0	82.4	60.4	61.6	82.7	62.7	63.0
	✓	✓			40.9	15.7	12.5	56.6	49.5	43.1	55.4	46.3	35.9
		✓	✓	✓	36.3	9.4	7.7	53.0	27.5	12.9	52.3	26.8	12.3
	✓	✓	✓		51.9	31.8	24.4	84.2	63.0	65.2	83.2	64.1	64.1
	✓	✓	✓	✓	52.3	31.6	24.5	84.4	63.2	65.4	84.0	65.3	65.5

Table 4. Ablation study under Non-IID missing patterns (%).

Dataset	Components				BRIDGE + MI			BRIDGE + DD			BRIDGE + CL		
	(a)	(b)	(c)	(d)	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
NUSWIDE	✓		✓		22.1	1.6	0.1	22.4	2.8	0.1	35.4	7.1	6.0
	✓	✓			24.6	2.5	0.4	22.1	1.8	0.1	25.2	3.5	0.9
		✓	✓	✓	23.7	2.0	0.7	29.9	4.8	3.2	30.2	6.9	5.0
	✓	✓	✓		30.1	4.6	3.6	35.6	6.3	5.2	37.8	8.8	7.5
	✓	✓	✓	✓	32.2	5.3	4.3	36.1	6.8	5.6	38.8	9.3	8.0
BDGP	✓		✓		45.0	13.0	11.5	37.3	8.5	6.9	53.8	21.9	19.1
	✓	✓			32.8	8.5	5.5	35.6	11.5	7.5	36.3	14.0	9.6
		✓	✓	✓	22.2	0.4	0.1	25.9	2.2	0.8	25.9	2.2	0.8
	✓	✓	✓		45.4	13.1	11.8	51.2	17.5	16.0	55.7	24.3	21.2
	✓	✓	✓	✓	45.5	13.2	11.8	52.4	18.5	17.3	56.6	24.8	22.3

patterns. We remove several key components and combine the remaining ones to obtain four methods for comparison. Specifically, (a) represents the module for training on reconstruction loss and on the complete data, such as contrastive learning, mutual information maximization, etc. (b) represents the distribution alignment module, as described in Section 2.2. (c) represents the missing view prediction module, as detailed in Section 2.3. (d) represents the weighted imputation method, as defined in Eq. (14). It should be noted that (d) must depend on the existence of both (b) and (c) (as it requires the results of distribution alignment and inter-view prediction), thus limiting additional possible combinations. We can observe that the method with complete components performs the best across all settings, regardless of the training method for complete data (i.e., MI, DD, CL), demonstrating the module's effectiveness. Removing (a) causes the model to lose its ability to represent the complete part, subsequently impacting the overall structure and leading to clustering collapse. The effects of (b) and (c) are equally notable, as their absence makes the model incapable of accurately predicting missing data, and the loss of only (d) also decreases clustering quality.

Additionally, Figure 4 presents a sensitivity analysis of parameter λ (defined in Eq. (14) on the BBCSport and Reuster ($\mathcal{M} = 0.5$). The change in λ affects the overall accuracy, and overly large or small values result in suboptimal clustering outcomes, confirming the validity of this module.

3.4. Model Analysis and Discussion

Beyond traditional metric comparison. We conduct an in-depth comparison of incomplete data processing modules using our newly proposed metric, FCR (see Eq. (17)). Specifically, we show the ratio of the final clustering accuracy across all data to the accuracy on complete data ($\mathcal{M} = 0.7$), as illustrated in Table 5. Evidently, FCR disregards the absolute magnitude of ACC and instead reflects the incomplete data processing module's capacity, and a higher FCR indicates superior corresponding capability of the model. BRIDGE+DD clearly achieves the best average performance, further highlighting the advantages of our method. Note that FCR is not necessarily strictly less than 1 (e.g., BRIDGE+DD achieves 109.32%), as the model may handle incomplete data better than complete ones.

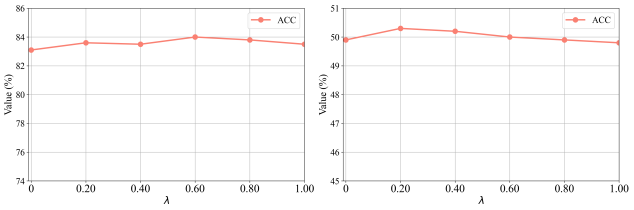


Figure 4. Parameter analysis: BBCSport (left) and Reuters (right).

Table 5. Comparison of Final-to-Complete Accuracy Ratio (FCR). The best/second-best results are highlighted in **bold**/underlined.

Method	BDGP	BBCSport	Reuters	NUSWIDE	Average
COMPLETER	83.49	87.12	94.71	89.38	88.68
DCP	88.55	93.59	93.82	94.45	92.60
DSIMVC	95.60	98.74	**101.78**	92.58	97.18
ProImp	83.98	102.51	-	-	93.25
CPSPAN	87.17	89.89	88.71	93.76	89.88
DIVIDE	95.55	95.21	90.80	97.62	94.80
BRIDGE+MI	**96.28**	94.15	87.82	**98.45**	94.18
BRIDGE+DD	94.39	**109.32**	98.18	92.75	**98.66**
BRIDGE+CL	95.65	96.28	92.69	88.16	93.20

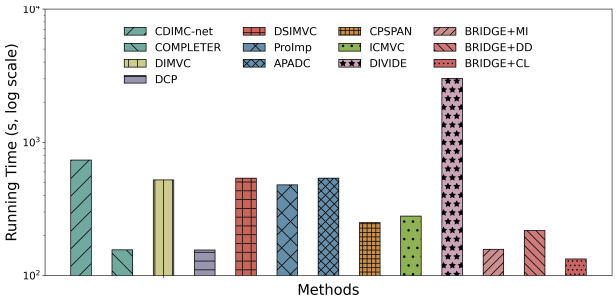


Figure 5. The runtime comparison of three instances of BRIDGE and ten baseline methods on Hdigit dataset.

Running time comparison. Figure 5 shows a comparison of the runtime between three instances of BRIDGE and ten baseline methods on Hdigit, where the missing rate is 0.7. We set the pre-training epochs to 500 for all models. Note that our methods has already accounted for the pretraining time on complete data. Overall, the runtime of BRIDGE is relatively short, as its classifiers, discriminators, and cross-view prediction models are lightweight networks, making it easier to deploy compared to more complex methods.

Interface flexibility verification. To thoroughly validate the

Table 6. Comparison results of BRIDGE combined with different DMVC methods, where 0.3, 0.5, and 0.7 represent missing rates.

Method	Metric	BDGP			BBCSport			NUSWIDE		
		0.3	0.5	0.7	0.3	0.5	0.7	0.3	0.5	0.7
Ours+MFLVC	ACC	95.5	93.7	92.4	92.7	87.1	63.0	57.5	53.4	46.3
	NMI	86.4	82.5	79.3	79.9	65.8	41.0	28.7	24.7	18.1
	ARI	89.2	85.2	82.1	81.6	70.3	33.3	26.8	22.7	16.0
	PUR	95.5	93.7	92.4	92.7	87.1	63.0	57.5	53.4	46.3
	FCR	96.4	96.7	95.1	102.1	100.7	104.8	92.3	90.6	92.0
Ours+CVCL	ACC	94.3	93.3	90.1	43.1	40.7	35.2	30.8	33.7	32.4
	NMI	83.4	81.1	74.9	23.8	15.0	13.1	10.1	11.3	10.3
	ARI	86.2	84.0	76.7	13.1	8.7	6.9	6.1	7.7	6.8
	PUR	94.3	93.3	90.1	54.4	40.7	40.9	49.4	45.7	46.8
	FCR	96.6	95.7	96.1	104.9	100.5	119.0	93.5	97.4	92.4
Ours+GCFAgg	ACC	95.6	92.9	92.2	78.0	80.2	70.7	54.7	53.0	50.4
	NMI	86.6	80.1	78.8	66.4	63.3	57.0	25.4	23.5	20.4
	ARI	89.4	83.2	81.8	60.2	61.4	57.2	23.4	21.5	18.8
	PUR	95.6	92.9	92.2	78.0	80.2	80.1	54.7	53.0	50.4
	FCR	97.0	96.0	94.9	101.7	101.3	99.3	93.6	90.3	90.4

flexibility of the interfaces included in BRIDGE, we select several DMVC methods, including MFLVC [40], GCFAggMVC [43], and CVCL [4], integrating BRIDGE with them to adapt to scenarios with incomplete data. Similar to the IMVC problem setup, we randomly remove a certain proportion of instances from these datasets, then input the complete data into the above three models for training, and subsequently connect the trained models to BRIDGE. Table 6 shows the results of various metrics, and we recommend paying close attention to FCR, as it reflects the quality of BRIDGE in bridging complete and incomplete data. We can see that FCR remains high across all settings, particularly in the BDGP and BBCSport, which is attributed to BRIDGE's excellent transferability. Furthermore, by comparing Tables 1 and 2, these methods also exhibit strong competitiveness. This demonstrates that BRIDGE is compatible with representation learning-based DMVC methods, enabling them to handle scenarios with incomplete data. This significantly expands the pool of DIMVC baselines.

Degree of convergence. We evaluate the convergence curves of BRIDGE+CL on Hdigit (left) and NUSWIDE (right) in Figure 6, where the missing rate is 0.7. Clearly, the convergence curve of BRIDGE+CL is very smooth with few fluctuations, and it reaches convergence within relatively few epochs, which is advantageous in practical applications.

Gain through distribution alignment. Besides being demonstrated in Tables (3)-(4), the effectiveness of the distribution alignment module is further validated through visualization results on four different datasets, as shown in Figure 7. We can intuitively observe that the features after distribution alignment exhibit a clearer clustering structure, with very uniform clusters and minimal inter-cluster adhesion, resulting in more distinct boundaries between clusters. That is, the clustering structure of features after distribution alignment is closer to label information, which will be more beneficial for downstream tasks.

Effectiveness of the classifier. Another advantage of BRIDGE is that it can utilize the classifiers (defined in Eq. (6)) directly for prediction. That is, the classifiers trained on complete data can be directly used to predict the labels of incomplete data, effectively transforming the clustering task into a classification task. Table 7 shows the clustering accuracy ACC_{clu} and classification accuracy ACC_{cls} across different datasets. Clearly, the two types of results vary in performance, with ACC_{clu} being better overall, but not significantly. In other words, if achieving a complete clustering structure without predicting missing data is acceptable, or if avoiding the overhead introduced by imputation is necessary, directly using ACC_{cls} is also a viable option.

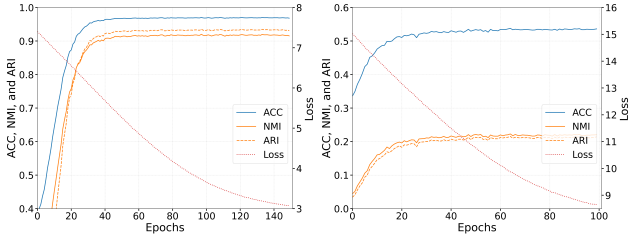
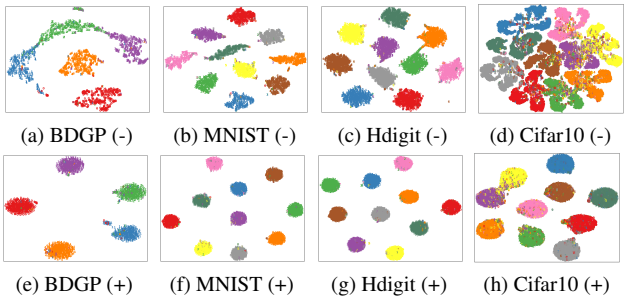


Figure 6. Convergence curves on Hdigit and NUSWIDE.



(a) BDGP (-) (b) MNIST (-) (c) Hdigit (-) (d) Cifar10 (-)

(e) BDGP (+) (f) MNIST (+) (g) Hdigit (+) (h) Cifar10 (+)

Figure 7. Visualization of the final clustering structure across different datasets using BRIDGE+CL, where (-) / (+) indicates the absence / presence of the distribution alignment module.

Table 7. Comparison of the overall accuracy of the classifiers and the final clustering accuracy on different datasets (%).

Metric	BDGP	BBCSport	Reuters	NUSWIDE	Hdigit
ACC_{clu}	93.44	75.96	46.17	53.78	97.02
ACC_{cls}	93.12	77.98	46.25	53.70	97.00

4. Conclusion

In this paper, we propose a novel framework BRIDGE, which is a unified interface for bridging from DMVC to DIMVC. Through our modular design and considering the Non-IID missing patterns, BRIDGE achieves good compatibility and fills the gap where existing methods do not account for the more complex Non-IID missing patterns in IMVC. Our method can be integrated into mainstream DMVC methods to improve their representation learning on incomplete multi-view data, and extensive experiments validated its effectiveness. We hope this methodology can facilitate the future work on establishing unified solutions for DMVC and DIMVC, as well as reducing their redundant work.

Acknowledgments

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